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BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT

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Department of Information Sciences and Engineering

Major Project - Phase 1 - Review 1

Automated Cybercrime Investigation Using AI and Multi-Agent Systems

UNDER THE GUIDANCE OF:

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Problem Statement

With the rapid increase in cyber crimes such as hacking, data breaches, and malware attacks, digital forensic investigation has become essential for identifying and analyzing digital evidence. However, current forensic investigation methods rely heavily on manual analysis and multiple independent tools to examine disk data, system logs, memory dumps, and network traffic. This process is time-consuming, complex, and requires expert knowledge to correlate evidence from different sources. As the volume of digital data continues to grow, there is a need for an automated and intelligent system that can efficiently analyze digital evidence, detect suspicious activities, and assist investigators in cybercrime investigation.



Motivation

Current Challenges

- Increasing use of AI-generated deepfake authentication attacks
- Supply chain poisoning in open-source software dependencies
- Multi-vector attacks that bypass isolated security tools

Limitations of Existing Systems

- Existing tools analyze disk, logs, memory, and network data separately.
- Lack of automation and intelligent evidence correlation.

Real-World Impact

- Faster cybercrime investigation
- Improved detection of suspicious activities
- Automated forensic report generation
- Better support for cybersecurity



Literature Survey(Table)

Title	Authors	Year	Algorithms	Dataset/Source Used	Results	Limitations
A Framework for Integrated Digital Forensic Investigation Employing AutoGen AI Agents markscanlon	Akila Wickramasekara, Mark Scanlon	2024	AutoGen agents, SW1H decomposition markscanlon	Hansen API prototypes markscanlon	Task automation, reusable subtasks markscanlon	Hallucinations, no large eval markscanlon
AutoGen Driven Multi-Agent Framework for Iterative Crime Data Analysis (LUCID-MA) arxiv	N/A (arXiv)	2025	AutoGen + LLaMA-2, iterative epochs arxiv	Chicago crime dataset arxiv	Prediction accuracy gains over epochs arxiv	Offline data only arxiv
AutoDFBench: AI-Generated DF Code Testing d-nb	Wickramasekara et al.	2025	Prompt eng., F1 on NIST d-nb	NIST CFTT (10k+ scenarios) d-nb	Claude 3.5: 0.421 score d-nb	Prompt-dependent perf. d-nb
Self-Organising Multi-Agent System for Decentralised Forensic Investigations kclpure.kcl.ac	N/A (Expert Systems)	2018	Self-organizing agents, coordination protocol kclpure.kcl.ac	Simulated networks kclpure.kcl.ac	Dynamic agent org., collective decisions kclpure.kcl.ac	Simulation-only kclpure.kcl.ac
Multi-Agent Systems for Dynamic Forensic Investigation kclpure.kcl.ac	N/A (ICIC)	2016	MAS model, network analysis agents kclpure.kcl.ac	Simulated cyber scenarios kclpure.kcl.ac	Malicious activity detection kclpure.kcl.ac	Narrow cyber focus kclpure.kcl.ac
Abstract Digital Forensics Model (ADFM) ijcrt	N/A (review)	2005	IDIP phases (ID, prep, approach, etc.) ijcrt	Generic DF cases ijcrt	Structured process model ijcrt	Static, no automation ijcrt
End-to-End Digital Investigation Process (EEDIP) ijcrt	N/A	2006	End-to-end phases incl. evidence mgmt. ijcrt	Cyber incident traces ijcrt	Comprehensive workflow ijcrt	Manual-heavy ijcrt
Integrated Digital Investigation Process (IDIP) ijcrt	N/A	2003	11 phases (traceback, dynamite, etc.) ijcrt	Early cyber crimes ijcrt	Holistic integration ijcrt	Rigid structure ijcrt
Digital Forensics Research Workshop Model (DFRWS) ijcrt	Pollitt	2003	Collection, exam, analysis, reporting ijcrt	NIST-like standards ijcrt	Standardized phases ijcrt	Lacks adaptability ijcrt
AI-Driven Digital Forensics ijsret	N/A	2024	ML/DL pattern recognition ijsret	Vast datasets (case studies) ijsret	Faster analysis, anomaly detection ijsret	Bias, explainability ijsret



Literature Survey(Summary)

- Difference between Existing and proposed system

1. Lack of Integrated Forensic Systems - Most existing research focuses on analyzing only one type of digital evidence, such as logs, network traffic, or disk data. They do not integrate multiple evidence sources into a single investigation framework.

Our Project: Integrates disk, memory, and log analysis together in one system.

2. Limited Automation - Many existing systems require manual investigation by forensic experts, even when AI is used for detection.

Our Project: Uses AI-based multi-agent automation to perform analysis automatically.

3. No Intelligent Evidence Correlation - Current research often analyzes evidence separately but does not correlate results to reconstruct attack timelines.

Our Project: Uses a correlation agent to combine results from multiple agents and identify attack patterns.

4. Lack of Multi-Agent Collaboration - Most existing forensic systems use single models or rule-based tools rather than collaborative intelligent agents.

Our Project: Implements a multi-agent framework using AutoGen, where each agent performs a specific forensic task.

5. Limited Automated Reporting - Many existing systems provide raw results that investigators must interpret manually.

Our Project: Automatically generates structured forensic investigation reports.



Research Gap Identified

Gap	Existing Failure	Your Improvement
Integrated Analysis	Single-source focus (logs/disk only)	Multi-agent fusion: disk/memory/log/network
Automation	Manual expert intervention persists	Full AutoGen workflow: acquisition to reporting Document1-3.pdf
Correlation	No timeline reconstruction across sources	Dedicated Correlation Agent links findings
Agent Collaboration	Single models, no teamwork	AutoGen multi-agents specialize & collaborate
Reporting	Raw outputs need manual interpretation	Automated PDF/JSON reports with timelines

Why Current Solutions Fail: Fragmented tools (Sleuth Kit, Volatility separate); no intelligent synthesis; expert-dependent; scale poorly

Scope: Our AutoGen system fills all gaps via specialized agents + correlation, enabling end-to-end automation



Proposed System

- Develop an **AI-based multi-agent digital forensic investigation system**.
- The system analyzes digital evidence such as:
 - Disk images
 - System logs
 - Memory dumps
 - Network traffic
- Each evidence type is processed by a **specialized forensic agent**.
- A **Correlation Agent** links findings from all agents to reconstruct the cyber attack timeline.
- The **Final Investigation Agent** generates an automated forensic investigation report.



Multi-Agent Digital Forensic System





Objectives

1. **Design a multi-agent framework** : for analyzing different types of digital forensic evidence.
2. **Correlate evidence from multiple sources** : to reconstruct cyber attack timelines.
3. **Generate automated forensic reports** : summarizing suspicious activities and investigation findings.



Methodology



INPUT

Investigator uploads
evidence files

 **Disk Image**
.img / .dd / .E01

 **Memory Dump**
.mem / .dmp / .raw

 **System Logs**
auth.log / syslog / .evtx

 **Network Capture**
.pcap / .pcapng

01 Data Collection

Collect & verify all evidence

 **Disk Image**
FTK Imager / dd command

 **Memory Dump**
DumpIt / Belkasoft

 **System Logs**
Win Event Viewer / /var/log

 **Network Capture**
Wireshark live capture

02 Preprocessing

Verify, clean & structure

 **SHA-256 Verify**
Hash before & after analysis

 **Metadata Extract**
Timestamps, user IDs, PIDs

 **Log Normalization**
Pandas filter + UTC align

 **JSON Structuring**
Standard schema per agent

03 Model & Algorithm

AI agents per evidence type

 **AutoGen LLM**
Multi-agent orchestration

 **Sleuth Kit**
Disk & file system forensics

 **Volatility**
RAM / memory forensics

 **Wireshark / Zeek**
Network packet analysis



Methodology

INPUT

01 Data Collection

02 Preprocessing

03 Model & Agents

04 Metrics

CORRELATION

OUTPUT

03 AI Agents

5 agents run in parallel



Evidence Agent

Collect & hash verify



Disk Agent

Sleuth Kit / Autopsy

04 Evaluation Metrics

How we measure success



Detection Accuracy

Target $\geq 85\%$



Time Reduction

Target $\geq 60\%$ faster



Correlation & Reasoning Agent

Combines all agent outputs — builds the complete attack timeline



Cross-source linking

Disk + Log + Memory + Network



Attack timeline build

Chronological event sequence



AutoGen Orchestrator — 5 Specialized Agents Working in Parallel

Evidence Acquisition Agent
SHA-256 · File I/O

Disk Analysis Agent
Sleuth Kit · Autopsy

Log Analysis Agent
Pandas · Regex · ELK

Memory Analysis Agent
Volatility Framework

Network Analysis Agent
Wireshark · Zeek/Bro



FINAL OUTPUT — Forensic Investigation Report



Tools and Technologies



Programming Languages

Python – Used for implementing AI agents, automation, and forensic data analysis.

JavaScript – Used for developing the web interface.

Frontend

React.js – Used to build an interactive user interface for uploading evidence and viewing results.

HTML5 & CSS3 – Used for designing the web interface.

Backend

Node.js – Used to handle server-side operations and requests.

Express.js – Used to create APIs and manage communication between components.

Framework

AutoGen – Multi-agent framework used to coordinate AI agents for forensic investigation tasks.

Software Tools

Sleuth Kit – Disk forensic analysis tool.

Autopsy – Tool for analyzing disk images and recovering files.

Volatility – Used for memory dump analysis.

Wireshark – Used for network traffic analysis.



Feasibility Study

- **Technical Feasibility**
- AutoGen, Python, Sleuth Kit, Volatility, Wireshark - all available & open-source
- Standard hardware (8GB RAM, multi-core) is sufficient to run the system
- GPT-4 / local LLM available via API for agent reasoning
- **Economic Feasibility**
- All major tools are open-source; zero licensing cost
- No expensive hardware or cloud infrastructure required
- Minimal development cost using existing frameworks
- High ROI- significantly reduces investigator time and effort
- **Operational Feasibility**
- Simple web dashboard, investigator uploads evidence, system returns report
- AI agents run in parallel, faster than manual tool-by-tool analysis
- Handles large datasets (disk images, logs, memory dumps) efficiently
- Auto-generates PDF / JSON / Markdown reports ready for legal use



SDG (Sustainable Development Goals) mapped (as per the metrics provided by Dept.)



This project aligns with the following **UN Sustainable Development Goals:**

- **SDG 9 - Industry, Innovation and Infrastructure**

Promoting innovation through AI-based forensic systems.

- **SDG 4 - Quality Education**

Preventing economic loss from cyber attacks.

- **SDG 16 - Peace, Justice and Strong Institutions**

Strengthening cybersecurity and cybercrime investigation.



Engineering Problem (WP)

- **WP (Evidence Analysis Automation) ↔ WK (Artificial Intelligence & Machine Learning):**

Applying AI techniques to automatically analyze large volumes of digital evidence and identify suspicious activities or abnormal system behavior.

- **WP(Multi-Source Evidence Correlation) ↔ WK (Data Analysis & Pattern Recognition):**

Using data processing and pattern recognition methods to correlate information from disk data, logs, memory dumps, and network traffic to reconstruct the sequence of cyber incidents

- **WP (Distributed Investigation Tasks) ↔ WK (Multi-Agent Systems):**

Designing a multi-agent architecture where different intelligent agents perform specialized forensic tasks such as disk analysis, log analysis, memory analysis, and network traffic monitoring.

- **WP (Cyber Attack Detection) ↔ WK (Cybersecurity & Digital Forensics):**

Applying digital forensic techniques and cybersecurity knowledge to detect malware activities, unauthorized access, and suspicious network communication.



Work Plan (Timeline)

•Project Phases

- Phase 1: Research & Planning** (*Month 1*) Review existing literature, finalize system architecture, define agent roles, set up development environment, identify datasets (NIST CFReDS, Digital Corpora).
- Phase 2: System Design** (*Month 2*) Design multi-agent architecture using AutoGen, design database schema, design UI wireframes for investigator dashboard, define API endpoints between agents and backend.
- Phase 3: Core Development** (*Month 3 - 4*) Implement all five AI agents (Evidence Acquisition, Disk, Log, Memory, Network), integrate forensic tools (Sleuth Kit, Volatility, Wireshark), develop preprocessing pipeline with SHA-256 verification and JSON structuring.
- Phase 4: Correlation & Reporting** (*Month 5*) Implement Correlation and Reasoning Agent using LLM, build attack timeline reconstruction logic, develop automated Report Generation Agent producing PDF, JSON, and Markdown outputs.



Work Plan (Timeline)

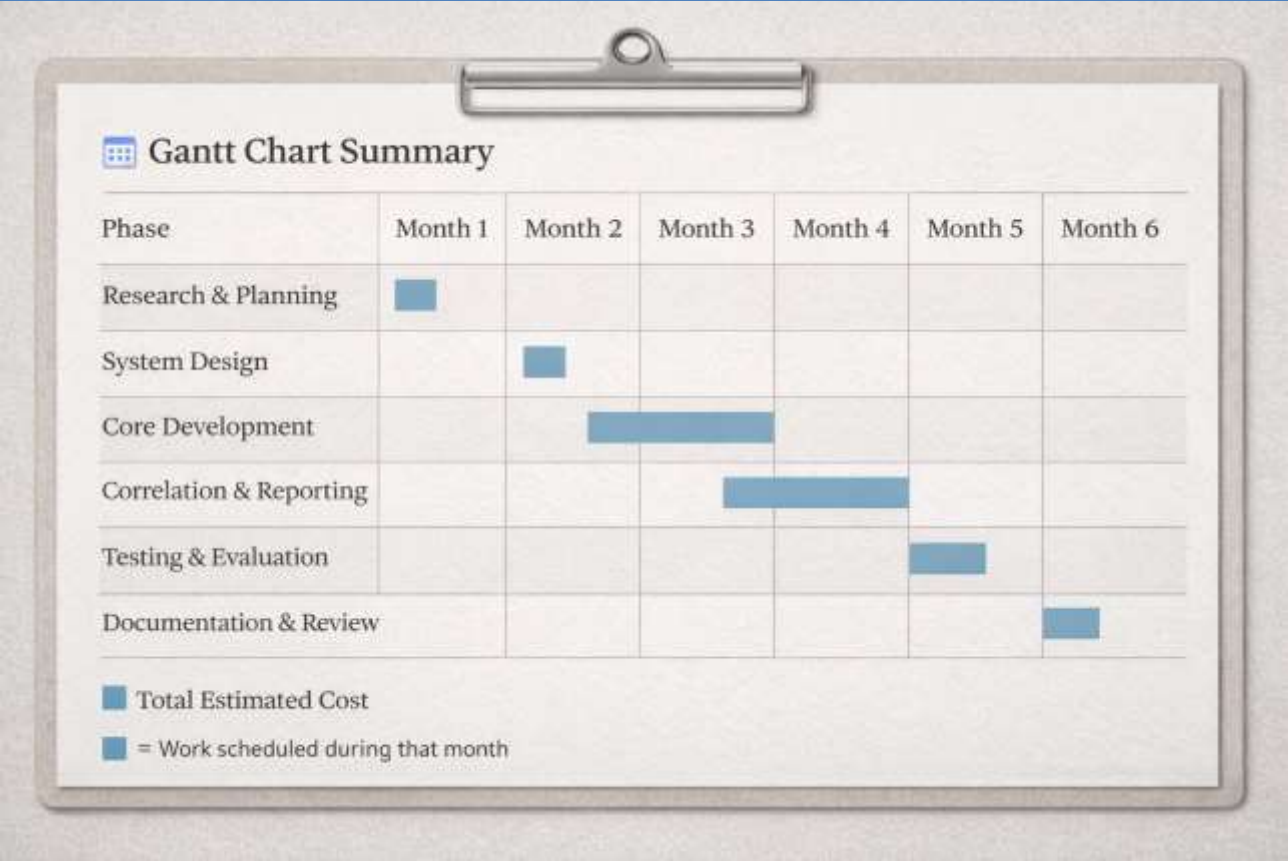
- Phase 5: Testing & Evaluation** (*Month 6*) Test system using forensic datasets, measure evaluation metrics (detection accuracy, time reduction, correlation efficiency, report quality), fix bugs and optimize performance.
- Phase 6: Documentation & Review** (*Month 7*) Write complete project documentation, prepare final report, conduct viva preparation, finalize presentation slides.

•Team Roles

Member	Role
Aishwarya Ramakrishna Hegde	Agent Development (Memory + Network Agent)
Deepthi P Kumar	Frontend Dashboard + Report Generation

Work Plan (Timeline)

- Gantt Chart



Budget Estimate

Item	Cost
Software & Tools	₹0 (all open-source)
Cloud / Hosting (optional)	₹0 – ₹2,000
Hardware (if needed)	₹0 (existing lab systems)
Dataset Access	₹0 (NIST CFReDS free)
Total Estimated Cost	₹0 – ₹2,000



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